

Scientific Computing & Advanced Programming Lab

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Lecture-Slides-6



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Gaussian Elimination

- We want to solve

$$A\mathbf{x} = \mathbf{b},$$

where

$$A \in \mathbb{R}^{n \times n}, \quad \mathbf{x}, \mathbf{b} \in \mathbb{R}^n.$$

- Write the system as an augmented matrix: $[A \mid \mathbf{b}]$.

Goal: Transform $[A \mid \mathbf{b}] \longrightarrow [U \mid \mathbf{c}]$, where U is upper triangular:

$$U = \begin{pmatrix} u_{11} & u_{12} & \cdots & u_{1n} \\ 0 & u_{22} & \cdots & u_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & u_{nn} \end{pmatrix}. \text{ Then solve } U\mathbf{x} = \mathbf{c}$$

using **back substitution**.



The Basic Elimination Step

- ▶ Suppose we are working with column k .
- ▶ The pivot is a_{kk} .
- ▶ For every row below the pivot i.e. $i = k + 1, \dots, n - 1$ calculate the multiplier: $m_{ik} = \frac{a_{ik}}{a_{kk}}$
- ▶ and perform the row operation: $R_i \leftarrow R_i - m_{ik}R_k$.
- ▶ This makes $a_{ik} = 0$.
- ▶ Entries below the pivot are eliminated.



Gaussian Elimination: Pseudocode

```
GaussianElimination(A, b)
  n = number of rows in A
  for k = 0 to n-2
    for i = k+1 to n-1
      m = A[i,k] / A[k,k]
      for j = k to n-1
        A[i,j] = A[i,j] - m*A[k,j]
      b[i] = b[i] - m*b[k]
  return A, b
```

The algorithm transforms

$$[A \mid \mathbf{b}] \longrightarrow [U \mid \mathbf{c}].$$



Back Substitution

After forward elimination we have

$$U\mathbf{x} = \mathbf{c}.$$

$$u_{11}x_1 + u_{12}x_2 + u_{13}x_3 = c_1,$$

For example, $u_{22}x_2 + u_{23}x_3 = c_2,$

$$u_{33}x_3 = c_3.$$

Start with the last equation and work upwards.

For $i = n - 1, n - 2, \dots, 0,$

$$x_i = \frac{c_i - \sum_{j=i+1}^{n-1} u_{ij}x_j}{u_{ii}}$$

$$x_{n-1} \rightarrow x_{n-2} \rightarrow \dots \rightarrow x_1 \rightarrow x_0$$



Back Substitution: Pseudocode

```
BackSubstitution(U, c)
  n = number of rows
  create x = zeros(n)
  for i = n-1 down to 0
    s = 0
    for j = i+1 to n-1
      s = s + U[i,j] * x[j]
    x[i] = (c[i] - s) / U[i,i]
  return x
```



Why Pivoting?

- ▶ In Gaussian elimination, we divide by the pivot: $m = \frac{A_{ik}}{A_{kk}}$.
- ▶ If $A_{kk} = 0$, the algorithm fails.
- ▶ try: Example:

$$\left[\begin{array}{cc|c} 0 & 1 & 1 \\ 1 & 1 & 2 \end{array} \right]$$

- ▶ The first pivot is zero. Swap the rows:

$$\left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 1 & 1 \end{array} \right].$$



Partial Pivoting

- ▶ At elimination step k , examine the current column:
 $A_{kk}, A_{k+1,k}, \dots, A_{n-1,k}$.
- ▶ Choose the entry with the largest absolute value:

$$p = \operatorname{argmax}_{i=k, \dots, n-1} |A_{ik}|$$

- ▶ Then swap:

$$R_k \leftrightarrow R_p.$$

- ▶ Example:

$$\begin{pmatrix} 0.001 \\ -5 \\ 2 \end{pmatrix} \longrightarrow \boxed{-5}$$

Choose the largest available pivot in the column.



Gaussian Elimination with Partial Pivoting

Input: A , b

n = number of rows

FOR $k = 0, \dots, n-2$:

 # Find pivot

p = index of largest $|A[i,k]|$,

$i = k, \dots, n-1$

 # Swap rows

 swap rows k and p of A

 swap $b[k]$ and $b[p]$

 # Eliminate below pivot

 # continue as usual



From Gaussian Elimination to LU

Gaussian elimination transforms

$$A\mathbf{x} = \mathbf{b}$$

into

$$U\mathbf{x} = \mathbf{c},$$

where U is upper triangular. But during elimination, we compute multipliers

$$m_{ik} = \frac{A_{ik}}{A_{kk}}.$$

Key idea: Instead of throwing these multipliers away, store them in a lower triangular matrix L .

$$A = LU$$



Where Does L Come From?

Consider one elimination step:

$$R_i \leftarrow R_i - m_{ik}R_k.$$

The multiplier

$$m_{ik} = \frac{A_{ik}}{A_{kk}}$$

is exactly what is needed to construct L . For example,

$$L = \begin{pmatrix} 1 & 0 & 0 \\ m_{21} & 1 & 0 \\ m_{31} & m_{32} & 1 \end{pmatrix}, \quad U = \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{pmatrix}.$$

Thus, Gaussian elimination gives us both:

U from the row operations

L from the multipliers



Why Factorize A ?

Once $A = LU$, the system $A\mathbf{x} = \mathbf{b}$ becomes

$$LU\mathbf{x} = \mathbf{b}.$$

Introduce

$$L\mathbf{y} = \mathbf{b}.$$

Then solve

$$U\mathbf{x} = \mathbf{y}.$$

So one system becomes two triangular systems:

$$A\mathbf{x} = \mathbf{b} \implies \begin{cases} L\mathbf{y} = \mathbf{b}, \\ U\mathbf{x} = \mathbf{y}. \end{cases}$$

Advantage: If A is fixed and many different $\mathbf{b}$'s are given, we factorize A only once.



LU Decomposition: Pseudocode

Input: Matrix A $U \leftarrow A$, $L \leftarrow I$

FOR $k = 0, \dots, n-2$:

FOR $i = k+1, \dots, n-1$:

$m = U[i, k] / U[k, k]$

$L[i, k] = m$

$U[i, k:] = U[i, k:] - m U[k, k:]$

RETURN L, U

$$A = LU$$



Using LU to Solve $A\mathbf{x} = \mathbf{b}$

Suppose

$$A = LU.$$

Then

$$A\mathbf{x} = \mathbf{b} \implies LU\mathbf{x} = \mathbf{b}.$$

Introduce an intermediate vector $\mathbf{y}$:

$$L\mathbf{y} = \mathbf{b}.$$

Then solve

$$U\mathbf{x} = \mathbf{y}.$$

$$L\mathbf{y} = \mathbf{b} \quad (\text{forward substitution})$$

$$U\mathbf{x} = \mathbf{y} \quad (\text{back substitution})$$



LU: Solving a System

Consider

$$A = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 5 \\ 11 \end{pmatrix}.$$

LU decomposition gives

$$L = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}.$$

Step 1: Forward substitution

$$L\mathbf{y} = \mathbf{b} \Rightarrow \begin{cases} y_1 = 5, \\ 2y_1 + y_2 = 11 \end{cases} \Rightarrow \mathbf{y} = \begin{pmatrix} 5 \\ 1 \end{pmatrix}.$$

Step 2: Back substitution

$$U\mathbf{x} = \mathbf{y} \Rightarrow \begin{cases} 2x_1 + x_2 = 5, \\ x_2 = 1 \end{cases}$$

